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**Category**
Lesson 1: Customary Units of Measurement

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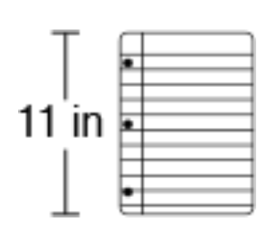
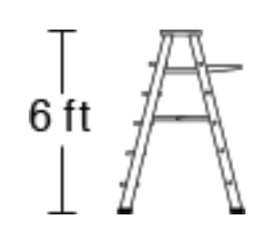
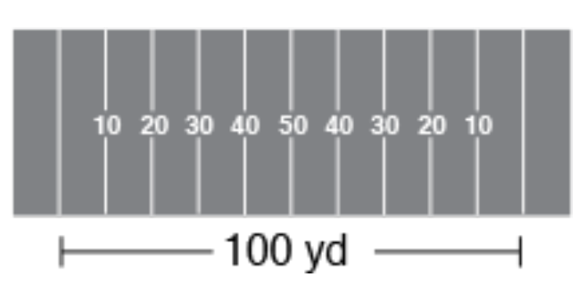
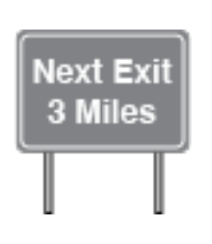
# Basic U.S. Customary Units

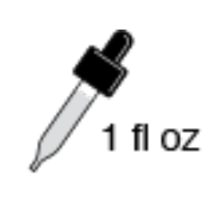
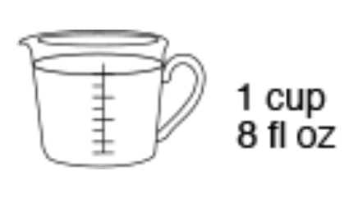
The measuring system most often used in the United States is called the **U.S. customary system.** The most often used customary units of length are shown below.

or distance

| Unit | Sample Use | Comparing Units |
|---|---|---|
| inch (in) |  | 1 in = $\frac{1}{12}$ ft |
| foot (ft) |  | 1 ft = 12 in<br>1 ft = $\frac{1}{3}$ yd |
| yard (yd) |  | 1 yd = 3 ft<br>1 yd = 36 in |
| mile (mi) |  | 1 mi = 1,760 yd<br>1 mi = 5,280 ft |

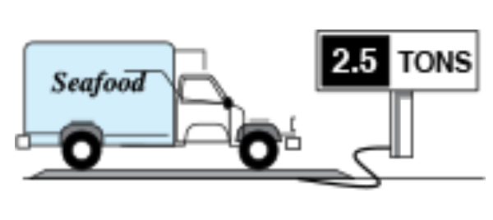
or volume

Units of capacity measure the space occupied by substances that can be poured (liquid or granular materials, for example). The most often used customary units of capacity are shown in the table.

| Unit | Sample Use | Comparing Units |
|---|---|---|
| fluid ounce (fl oz) |  | 1 fl oz = $\frac{1}{8}$ c |
| cup (c) |  | 1 c = 8 fl oz |
| pint (pt) | | 1 pt = 2 c |
| quart (qt) | | 1 qt = 2 pt |
| gallon (gal) | | 1 gal = 4 qt |

force of gravity

Weight is a measure of how heavy an object is. In the customary system, weight is measured in the three basic units shown in the table.

| Unit | Sample Use | Comparing Units |
|---|---|---|
| **ounce (oz)** | | **1 oz = $\frac{1}{16}$ lb** |
| **pound (lb)** | | **1 lb = 16 oz** |
| **ton (t)** |  | **1 t = 2,000 lb** |

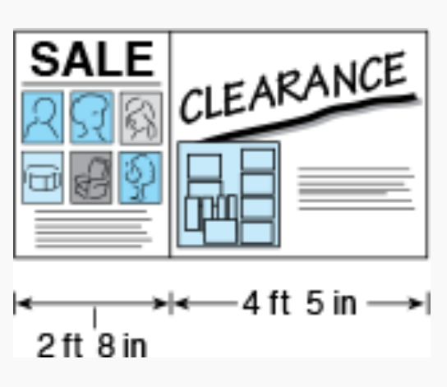
or tonne

When working with length, add or subtract each unit separately. With addition and subtraction, estimation helps you know what to expect. If you quickly add or subtract the larger units first, you can tell if your answer makes sense. However, estimation isn't always useful when you multiply or divide.

### Example 1

Hide

The manager of Go-Mart wants a display measuring 4 ft 5 in wide placed next to a display measuring 2 ft 8 in wide. How much space will the two displays take up?



**Step 1**
Set up the problem.

$$\begin{array}{r} 4\text{ ft } 5\text{ in} \\ +2\text{ ft } 8\text{ in} \\ \hline \end{array}$$

**Step 2**
Estimate first by adding the larger units.
4 ft + 2 ft = 6 ft
The answer should be slightly more than 6 ft.

**Step 3**
Add the units separately.

$$\begin{array}{r} 4\text{ ft } 5\text{ in} \\ +\ 2\text{ ft } 8\text{ in} \\ \hline 6\text{ ft } 13\text{ in} \end{array}$$

but 1 ft = 12 inches

**Step 4**
Regroup the smaller unit and combine with the larger unit, if necessary.
6 ft + 13 in = 6 ft + 1 ft + 1 in = **7 ft 1 in**
7 ft 1 in is slightly more than 6 ft. The two displays will take up **7 ft 1 in.**

### Example 2

Hide

The owners of Go-Mart want to install a 12-ft-long cooler. The space they have in mind is 10 ft 8 in long. How much more space is needed for the cooler to fit?

**Step 1**
Set up the problem.

$$\begin{array}{r} 12\text{ ft} \\ -10\text{ ft } 8\text{ in} \\ \hline \end{array}$$

**Step 2**
Estimate first by subtracting the larger units.
12 ft − 10 ft = 2 ft
The answer should be slightly less than 2 ft.

**Step 3**
Regroup from the larger unit, if necessary.
12 ft = 11 ft 12 in

**Step 4**
Subtract each unit separately.

$$\begin{array}{r} 11\text{ ft } 12\text{ in} \\ -\ 10\text{ ft } 8\text{ in} \\ \hline 1\text{ ft } 4\text{ in} \end{array}$$

1 ft 4 in is slightly less than 2 ft. Go-Mart needs **1 ft 4 in** more space.

### Example 3

Hide

Anna needs a wooden dowel that she can cut into 5 pieces. Each piece must be 1 ft 3 in. How long should the wooden dowel be?

**Step 1**
Set up the problem.

$$\begin{array}{r} 1\text{ ft } 3\text{ in} \\ \times\qquad 5 \\ \hline \end{array}$$

**Step 2**
Multiply the number of each unit separately.

$$\begin{array}{r} 1\text{ ft } 3\text{ in} \\ \times\qquad 5 \\ \hline 5\text{ ft } 15\text{ in} \end{array}$$

**Step 3**
Simplify, if possible.
5 ft 15 in = 5 ft + 1 ft + 3 in = 6 ft 3 in
The wooden dowel should be **6 ft 3 in.**

### Example 4

Hide

If you divide a 13-ft-3-in board into 3 equal pieces, how long is each piece?

**Step 1**
Set up the problem.

$$3\overline{)13\text{ ft } 3\text{ in}}$$

**Step 2**
Divide the larger unit.

$$\begin{array}{r} 4 \\ 3\overline{)13\text{ft}} \\ -12 \\ \hline 1 \end{array}$$

long division

Remainder: 1 ft = 12 in

**Step 3**
Regroup and add any remainder to the smaller unit. Divide.
12 in + 3 in = 15 in

$$\begin{array}{r} 5 \\ 3\overline{)15\text{in}} \\ -15 \\ \hline 0 \end{array}$$

→ remainder

**Step 4**
Combine the amounts from Steps 2 and 3 to get the answer.
4 ft + 5 in = 4 ft 5 in

Each piece should be **4 ft 5 in** long.

**Tip**

Show

*When dividing, you can estimate using compatible numbers. In this problem, 13 isn't divisible by 3, but 12 is. 12 ft ÷ 3 is 4 ft. The answer should be close to 4 ft.*

### Key Terms

- **U.S. customary units of measurement**
- **rate**
- **unit multiplier**

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